Feature-Importance Report

Purchase Prediction Model — E-Commerce Mini-Project 5
Model: Optimized Logistic Regression (standardized coefficients)

Top 10 Influential Features

#	Feature	Coefficient	Direction
1	Time on site	1.25	Positive
2	Days since last visit	-0.31	Negative
3	Discount used	0.23	Positive
4	Previous purchases	0.18	Positive
5	Cart items	0.17	Positive
6	Pages viewed	0.15	Positive
7	Review score viewed	0.15	Positive
8	Session count	0.14	Positive
9	Location: West	-0.10	Negative
10	Email clicked	0.09	Positive

Business Interpretation & Recommended Actions

Feature	Interpretation	Recommended Action
Time on site	Longer engagement signals stronger buying intent	Trigger on-site prompts (chat, urgency messaging) once a visitor crosses an engagement threshold
Days since last visit	Interest decays the longer a customer stays away	Automate re-engagement email/SMS once recency passes a threshold
Discount used	Price-sensitive customers respond to discounts	Target discount offers at price-sensitive segments identified by prior discount use
Previous purchases	Repeat customers are more likely to purchase again	Invest in loyalty and retention programs for existing customers

Cart items	More items in cart signals stronger purchase intent	Send timely cart-abandonment reminders
Pages viewed	Deeper browsing indicates active product research	Surface personalized product recommendations as page views increase
Review score viewed	Customers viewing higher-rated products convert more	Highlight top-rated products in recommendation widgets
Session count	Returning across multiple sessions signals sustained interest	Retarget multi-session visitors who haven't yet purchased
Location: West	West-region customers convert somewhat less often	Investigate regional fulfillment, pricing, or localization gaps
Email clicked	Email engagement is associated with higher purchase odds	Continue investing in targeted email campaigns

Note: Coefficients reflect standardized association with the model's predicted log-odds of purchase, not proven causation. Positive values raise predicted purchase likelihood; negative values lower it.